

Yifei Jiao

<https://yifei-jiao.github.io>

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ABOUT ME

Hello! I am Yifei Jiao (he/him), a postdoc researcher at University of California, Santa Cruz. My research interests include **collisional/tidal induced dynamical evolution of small bodies, rings and satellites**. I would enjoy exploring any unsolved questions of our Solar System and beyond with theoretical and numerical methods.

EDUCATION

Tsinghua University, Beijing, China <i>PhD with Prof. Hexi Baoyin</i> <i>Numerical modeling of impact dynamics and collisional evolution of asteroids</i>	Sep 2020–Jun 2025
Tsinghua University, Beijing, China <i>Bachelor of Engineering</i> <i>Monte Carlo tree search for designing multi-target deep space exploration</i>	Sep 2016–Jun 2020

EXPERIENCE

University of California Santa Cruz, CA, USA <i>Postdoc with Prof. Francis Nimmo</i>	Sep 2025–now
Shanghai Astronomical Observatory, Shanghai, China <i>Visiting Scholar with Prof. Xian Shi</i>	Aug 2025
University of Tokyo & NAOJ, Tokyo, Japan <i>Visiting Scholar with Prof. Seiji Sugita</i>	May 2025
LPL, University of Arizona, Tucson, AZ, USA <i>Visiting Student with Prof. Erik Asphaug</i>	Jan–Jun 2024
Tsinghua University, Beijing, China <i>Teaching Assistant for Theoretical Mechanics</i>	2020, 2021

GRANTS

National Natural Science Foundation of China (¥ 300k) <i>Student PI</i> <i>Impact dynamics and collisional evolution of highly porous asteroids</i>	2024–2025
Youth Talent Support Program of China Association for Science and Technology (¥ 40k) <i>Student PI</i>	2024–2025

HONORS AND AWARDS

Asteroid Name (20765) YIFEIJIAO	2026
Best Poster Award of Asteroids, Comets, Meteors Conference	2026
Excellent Doctoral Dissertation Award of Chinese Society of Astronautics	2025
Excellent Doctoral Dissertation Award of Tsinghua University	2025
Last updated: July 2026	

Excellent Doctoral Graduate Award of Beijing Municipality	2025
Tsinghua Top-Grade Scholarship (Highest Honor for Students, 10 per year)	2024
China National Scholarship	2024
Excellent Paper Award of Young Scientist Forum of Planetary Science, China	2023
Tsinghua Short-term Study Abroad Scholarship	2023
Tsinghua First/Second Prize Scholarship	2022, 2023
China Trajectory Optimization Competition, 2nd Place	2020
Tsinghua Excellent Technology & Innovation Scholarship	2019
Tsinghua Excellent Academic Scholarship	2017, 2018, 2019

PUBLICATIONS

(7 first-author, 16 total, *=mentee/student)

1. **Y. Jiao**, F. Nimmo, J. Wisdom, R. Dbouk. Elliptical impact craters on Saturnian moons as evidence of a recent tidal disruption event. (*in prep*)
2. Y. Huang, **Y. Jiao**, N. Torii, W. Dai, E. Kokubo, T. Ito. Tidal disruption of ancient Plutos as the origin of trans Neptunian rebels. *Science* (*submitted*)
3. **Y. Jiao**, F. Nimmo, J. Wisdom, R. Dbouk. Investigating tidal stripping of a pre-existing moon as the origin of Saturn's young icy rings. *Astrophys. J. Lett.* (2026)
4. **Y. Jiao**, B. Cheng, W. Dai, E. Asphaug, et al. Giant craters on asteroid Mathilde revealing the cohesive porous interior of carbonaceous parent bodies. *Nat. Geosci.* (*under review*)
5. **Y. Jiao**, B. Cheng, H. Baoyin. Probing the Moon from future asteroid impacts: a review. *Sci. China Technol. Sci.* (2026)
6. Y. Wu*, **Y. Jiao**, W. Dai, Y. Huang, Z. Liu, B. Cheng, H. Baoyin, J. Li. Detectability of lunar-origin asteroids in the LSST era. *Astrophys. J.* (2026)
7. Y. He*, Y. Wu*, **Y. Jiao**, W. Dai, X. Liu, B. Cheng, H. Baoyin. Observation timelines for the potential lunar impact of asteroid 2024 YR4. *Astrophys. J.* (2026)
8. Y. Wu*, Y. Li, B. Zheng, ..., **Y. Jiao**, et al. The START mission: probing the 2029 Apophis flyby via a low-cost geocentric smallsat. *Planet. Sci. J.* (*under review*)
9. W. Dai, B. Cheng, Y. Huang, **Y. Jiao**, et al. Diverse configurations of binary asteroids explained by multi-generation satellites. *Nat. Commun.* (2026)
10. X. Yan, P. Michel, R. Ni, **Y. Jiao**, J. Li. The Material Point Method (MPM) for simulating hypervelocity impact on asteroids. *Icarus* (2026)
11. T. Baoyin*, **Y. Jiao**, B. Cheng. Predicting the collision history of basaltic asteroids from parametrized shapes with an artificial neural network. *Mon. Not. R. Astron. Soc.* (2025)
12. **Y. Jiao**, B. Cheng, Y. Huang, E. Asphaug, et al. Asteroid Kamo'oalewa's journey from the lunar Giordano Bruno crater to Earth 1:1 resonance. *Nat. Astron.* (2024)
13. **Y. Jiao**, X. Yan, B. Cheng, H. Baoyin. SPH-DEM modeling of hypervelocity impacts on rubble-pile asteroids. *Mon. Not. R. Astron. Soc.* (2023)

14. N. Zhang, Z. Zhang, **Y. Jiao**, H. Baoyin. Multi-trajectory combination for multiple ground target observation by maneuvering on-orbit satellites. *IEEE Trans. Aerosp. Electron. Syst.* (2023)
15. **Y. Jiao**, B. Cheng, H. Baoyin. Optimal kinetic-impact geometry for asteroid deflection exploiting Delta-V hodograph. *J. Guid. Control Dyn.* (2022)
16. Z. Zhang, N. Zhang, **Y. Jiao**, H. Baoyin, J. Li. Multitree search for multisatellite responsiveness scheduling considering orbital maneuvering. *IEEE Trans. Aerosp. Electron. Syst.* (2021)

CONFERENCES/SEMINARS

Seminar, Southwest Research Institute, Boulder, CO, USA	Aug 2026
Asteroids, Comets, Meteors Conference, Poznan, Poland	Jul 2026
Lunar and Planetary Science Conference, The Woodlands, TX, USA	Mar 2026
Seminar, Shanghai Astronomical Observatory, Shanghai, China	Aug 2025
PhD Academic Forum of Tsinghua University, Beijing, China	May 2025
Seminar, National Astronomical Observatory of Japan, Tokyo, Japan	May 2025
Seminar, University of Tokyo, Tokyo, Japan	May 2025
Europlanet Science Congress, Berlin, Germany	Sep 2024
Young Scientist Forum of Planetary Science, Sanya, China	Mar 2023

SOFTWARE

1. **Y. Jiao**, et al. The SPHSOL code, which is a parallel smooth particle hydrodynamics C++ solver for simulating hypervelocity impact and tidal response of planetary bodies. <https://sphsol-tutorial.readthedocs.io>

STUDENT MENTORING

Yixuan Wu (Tsinghua University)	2025
Yifan He (Tsinghua University)	2025
Tamier Baoyin (Dartmouth College)	2024

SERVICES

Referee for Nat. Commun., Space Sci. Technol., Earth Planet. Phys.

Co-founder of Student-led Threatening Asteroid Reconnaissance of Tsinghua

Co-organizer of UCSC Earth & Planetary Science IGPP Seminar 2026